

# Manual

## Controlling Machine Data MDE-CMD 8.2

Version 1.1.23049

Last changed on: 01.09.2020

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# 1 Overview: Machine Data Controlling

## Purpose

Machine Data Controlling provides the ability to evaluate data collected in the system as concerns how recorded durations are distributed to various time accounts. It also analyzes any statuses that occurred with respect to the duration and number of occurrences.

## Implementation notes

You use the function package if:

- You would like to assess downtimes / statuses of separate machines or machine groups over selected time periods.
- You would like to evaluate status classes for a selected machine or group of machines over selected time periods or shifts.
- You would like to evaluate the time accounts of a selected machine or group of machines over selected time periods or shifts.

## Integration

The system accesses the times and quantities entered and statuses posted in MDE to display data.

## Features

- Status report (workplace/machine-related)
  - Tabular downtime / status assessment of separate machines over selected time periods. Graphical comparison of production times and downtimes. Graphical presentation of a machine ranking list showing the machines with either the longest or the most downtimes.
- Status report
  - Tabular downtime / status evaluation of a selected machine or group of machines across selected time periods or shifts. Graphical comparison of production times and downtimes. Graphical presentation of a status ranking list showing the longest or the most downtimes.
- Status class report
  - Tabular and graphical status class evaluation of a selected machine or group of machines over selected time periods or shifts.
- RPA report
  - Tabular and graphical RPA evaluation of a selected machine or group of machines over selected time periods or shifts.

- Status profile
  - Chronological evaluations of status / downtimes of separate machines or machine groups. Presentation of data in table and pivot table form. Data evaluation in pivot table based on various criteria. Graphical data presentation in pivot table form.
- Status class profile
  - Chronological evaluations of status classes for separate machines or machine groups. Presentation of data in table and pivot table form. Data evaluation in pivot table based on various criteria. Graphical presentation of data in pivot table form.
- RPA profile
  - Evaluation of resource performance accounts of separate machines or machine groups in tabular and graphical form.
- Status analysis
  - Evaluation of status / downtimes using a pivot table. Definition of the pivot table presentation based on various criteria. Graphical presentation (bar chart).
- Status hit list
  - Table showing cumulative times over a time period for active machine statuses, each sorted by duration and restricted to the x longest durations.
- Minor/major stops
  - Table showing cumulative times over a time period for active workplace / machine statuses, broken down into short (minor) and long (major) statuses.
- ABC analysis
  - Table showing cumulative times over a time period for active workplace / machine statuses, categorized into three ABC classes, each depending on the relationship of the durations in percent. Drill-down option used to list separate status times in a detail table.

## 2 Status Report, Machine-Related

### Summary

|                        |                                                                                     |
|------------------------|-------------------------------------------------------------------------------------|
| Menu                   | Operating facilities management → Status analyses → Status report (machine-related) |
| Transaction code       | mstrpm                                                                              |
| Function authorization | mstrpm                                                                              |

### Usage

The status report (machine-related) is an evaluation in operating facilities management. The analysis provides summarized status information relating to a workplace/ machine for a certain period of time and a certain number of workplaces. The data is presented in charts for informational purposes. You can also compare several machines with each other using this status report.

### Selection criteria

The application provides the following selection criteria:

#### Workplace

Search by machine/ by workplace. You can also run a search using wildcards.

#### Group

Search by workplaces/ machines that are assigned to the group that was entered. The selection is made using the field Group in the workplace/ machine configuration. You can also run a search using wildcards.

#### Cost center

Search by workplaces/ machines that are assigned to the cost center that was entered. You can also run a search using wildcards.

#### Company

Search by workplaces/ machines that are assigned to the company that was entered. You can also run a search using wildcards.

#### Responsibility area

This selection criterion references the responsibility area in the machine master data. Keep in mind that only machines are displayed that the user has also assigned responsibility areas to.

### Report group

This selection criterion references the report groups. All machines or workplaces are displayed that are assigned to the selected report group.

### Short designation

This selection criterion references the short name of the machines in the master data. All of the machines or workplaces are displayed that match the string that was entered. You can also run a search using wildcards.

### Designation

This field references the short name of the machines and workplaces in the machine's master data. At the bottom, only those machines are displayed that are identical to the string that was entered. You can also run a search using wildcards (placeholders \*) in this field.

### Inclusive status for RPA 11

Accounts for the status for RPA 11 (usually, this is the "Production" status).

### Date

The period of time from which data should be selected.

When selecting via shift(s), the shift date is evaluated, while when selecting by time the selection is based on the start date. Please keep in mind that a selection by shift is only supported for ADE and MDE data, not for WRM data.

### Shift(s)/ time

Selection by shift (only ADE and MDE events) or by time period. If no shift has been selected, all shifts are considered.

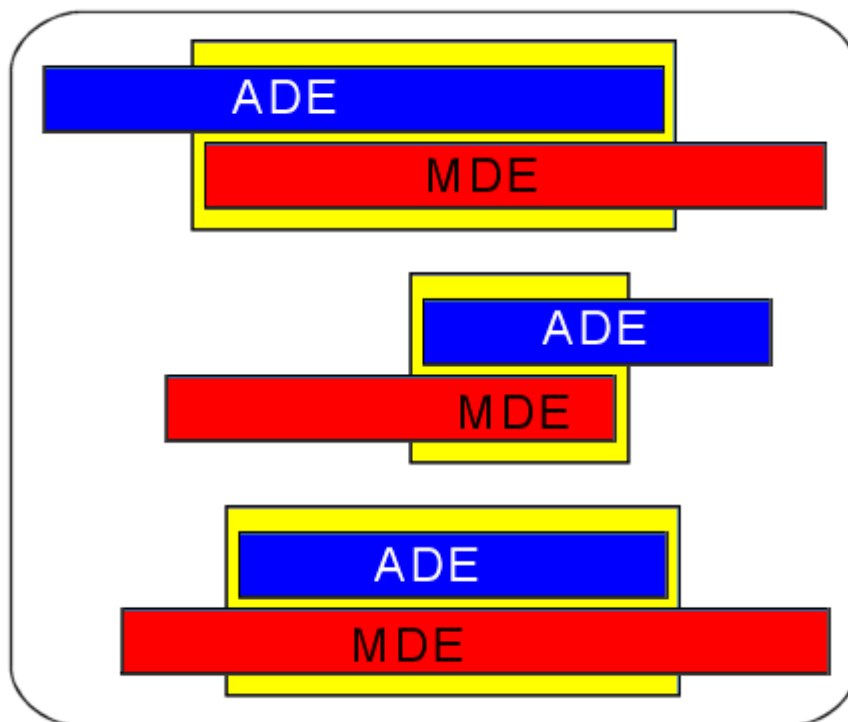
The two times each refer to the start or to the end of the date periods listed above.

### Order/ article



For this kind of evaluation type, only the finished ADE postings are considered. If the order is currently still running on the machine, the time period between the last logon and now is not taken into account. As such, it is by all means possible that there are differences between the machine evaluation and the order-related evaluation. Only ADE postings are taken into account that have started during the evaluation period. If necessary, the selection period must be selected so that the ADE postings that are to be taken into account are within this selection period. For this order-related evaluation, MPDV recommends the shift-related selection option.

The following illustration shows an example of an overlapping of ADE and MDE postings. The ADE postings take priority in this evaluation type. The yellow areas illustrate the results of this evaluation. MDE quantities and durations are calculated on a pro rata basis to achieve the result.



If orders are logged on in parallel at the machine, for this evaluation type, the full machine time (yellow area) and quantity is assigned to each order. The fact that orders are run in parallel will not result in a proportionate calculation.

### Resource type/ resource

For this kind of evaluation type, here again only the finished postings are considered. For this evaluation type, only the resource postings take priority. The principle is the same as when running an evaluation by order.

### Additional selection notes

#### Long term data

If the selection period exceeds the period for the online data area, the system applies the implicit solution and selects the medium-term data area as well. Therefore, there is no need for an explicit activation in order to be able to access the medium-term data set.

#### Determining a shift-adjusted quantity

This option known from MDE 7.2 console is set by default in MOC. What this means is that postings that were generated as a result of a shift change are not considered when determining the quantity. All that is accounted for here is the exact moment when a machine status was set. If this moment is outside of the evaluation interval, in this case the output will be 0.



## Status report - machine related detail application

In the tabular evaluation, the detailed status of a workplace/ a machine or group is presented within an arbitrarily selected period of time. The durations and quantities (frequencies) are compressed to each separate status.

The following data is available:

### ***Machine/ workplace category***

#### **Workplace**

Number of the machine/ workplace.

#### **Group**

Group that the workplace is assigned to based on the resource configuration.

#### **Cost center**

Cost center that the workplace is assigned to based on the resource configuration.

#### **Company**

Company that the workplace is assigned to based on the resource configuration.

### ***Duration category***

#### **Duration**

Total time of all disturbances within the evaluation period (i.e. not all downtimes within this evaluation period are compressed in this field).

%

Proportion of total downtime durations for the specific machine in relation to the total duration of disturbances of all of the displayed workplaces shown in percent.

#### **Production**

Durations posted to RPA 11

#### **RPA**

When delivered from the factory, the durations set for the resource performance accounts 1-10 are listed in the pool of columns. They are stored there with their abbreviation and can be shown as needed.

Please keep in mind that there are resource performance account columns for duration and quantities.

### ***Quantity category***

#### **Quantity**

Total number of disturbances that was active at each machine within the evaluation period.

%

Proportion of the number of downtimes for the specific machine in relation to the total number of all of the workplaces displayed shown in percent.

### **Production**

Number of RPA 11 statuses in the evaluation period.

### **RPA**

When delivered from the factory, the number of resource performance accounts 1-10 is listed in the pool of columns. They are stored there with their abbreviation and can be shown as needed.

Please keep in mind that there are resource performance account columns for duration and quantities.

### **Rate of capacity utilization**

Rate of capacity utilization in %

## **Comparison production - downtimes detail application**

The "Comparison: production - downtimes" detail application provides a summary of durations for production (RPA 11) and for the downtimes (RPA 1-10) for the workplaces marked in the table and these are compared graphically in the form of a bar chart.

The percentages shown relate to the total of both bars.

## **Machine hit list (durations) detail application**

The downtime durations for each of the workplaces marked in the table are shown in the form of a bar chart in the machine hit list (durations) detail application.

In the combo box "Displayed series" you can define whether the durations displayed as a percentage should relate to the total number of downtimes (RPA 1-10) or to the total duration (RPA 1-11). If both options have been selected, both bars are displayed (for each workplace).

## **Machine hit list (quantity) detail application**

In the machine hit list (quantity) detail application, the number of downtimes per machine is displayed in a chart. As opposed to the previous detail applications, no further display options are available here.

## 3 Status Report

### Summary

|                        |                                                                  |
|------------------------|------------------------------------------------------------------|
| Menu                   | Production Facility Management → Status analyses → Status report |
| Transaction code       | mstrp                                                            |
| Function authorization | mstrp                                                            |

The status report is an indispensable tool for any production executive. Whether planner, foreman or team leader, this report can be individually configured to show downtimes in any form.

### Usage

This evaluation provides status information relating to a workplace/ machine for a specific period of time and a certain number of workplaces. All of the accrued downtimes are pulled together in this evaluation and can be prepared in the form of a graphic or in table form based on what the user intends to achieve.

### Selection criteria

The application provides the following selection criteria:

#### Workplace

Search by machine/ by workplace. You can run a search using wildcards.

#### Group

Search by workplaces/ machines that are assigned to the group that was entered. The selection is made using the field Group in the workplace/ machine configuration. You can run a search using wildcards.

#### Cost center

Search by workplaces/ machines that are assigned to the cost center that was entered. You can run a search using wildcards.

#### Company

Search by workplaces/ machines that are assigned to the company that was entered. You can run a search using wildcards.

#### Responsibility area

This selection criterion references the responsibility area in the machine master data. Keep in mind that only machines are displayed that the user has also assigned responsibility areas to.

**Report group**

This selection criterion references the report groups. All machines or workplaces are displayed that are assigned to the selected report group.

**Short designation**

This selection criterion references the short name of the machines in the master data. All of the machines or workplaces are displayed that match the string that was entered. You can also run a search using wildcards.

**Designation**

This selection criterion references the name of the machines and workplaces in the machine's master data. At the bottom, only those machines are displayed that are identical to the string that was entered. You can also run a search using wildcards (placeholders \*) in this field.

**Status**

Limits the selection to a certain status. Displayed in the combo box are the status texts that the selection is filtered through.

**Include status for RPA 11**

Accounts for the status for RPA 11 (usually, this is the "Production" status).

**Date**

The period of time from which data should be selected.

When selecting via shift(s), the shift date is evaluated, whereas when selecting by time, the selection is based on the start date. Please keep in mind that a selection by shift is only supported for ADE and MDE data, not for WRM data.

**Shift/ time**

Selection by shift (only ADE and MDE events) or by time period. If no shift has been selected, all shifts are considered.

The two times each refer to the start or to the end of the date periods listed above.

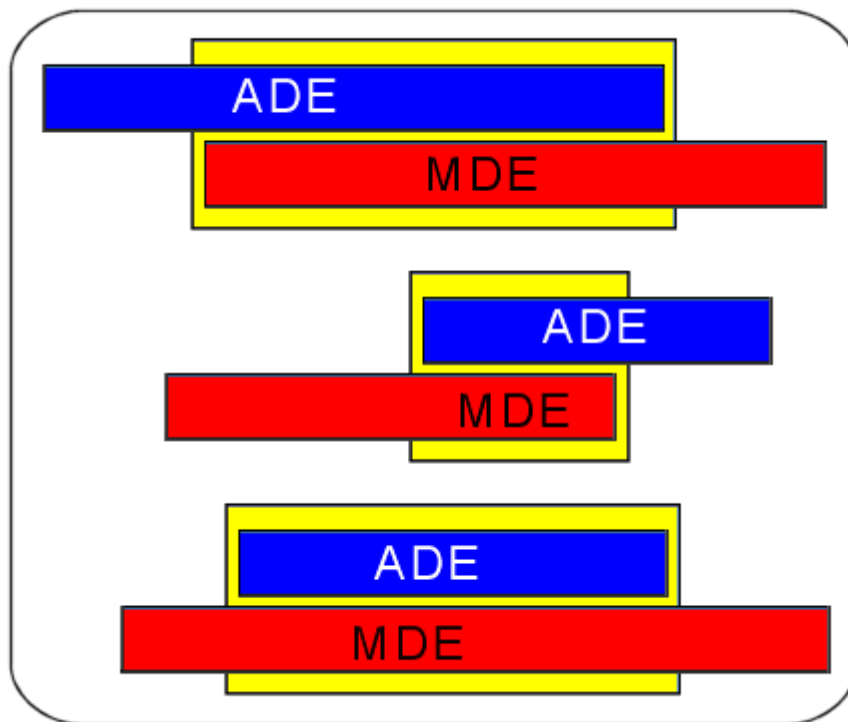
**Order/ article**

When you choose the option "Order", you must enter an order or an article.



For this kind of evaluation type, only the finished ADE postings are considered. If the order is currently still running on the machine, the time period between the last logon and "now" is not taken into account. As such, it is by all means possible that there are differences between the machine evaluation and the order-related evaluation. Only ADE postings are taken into account that have started during the evaluation period. If necessary, the selection period must be selected so that the ADE postings that are to be taken into account are within this selection period. For this order-related evaluation, MPDV recommends the shift-related selection option.

The following illustration shows an example of an overlapping of ADE and MDE postings. The ADE postings take priority in this evaluation type. The yellow areas illustrate the results of this evaluation. MDE quantities and durations are calculated on a pro rata basis to achieve the result.



If orders are logged on in parallel at the machine, for this evaluation type, the full machine time (yellow area) and quantity is assigned to each order. The fact that orders are run in parallel will not result in a proportionate calculation.

#### Resource type/ resource

When choosing the option "Resource", a resource must be entered.

For this kind of evaluation type, here again only the finished postings are considered. For this evaluation type, only the resource postings take priority. The principle is the same as when running an evaluation by order.

#### Additional selection notes

##### Long term data

If the selection period exceeds the period for the online data area, the system applies the implicit solution and selects the medium-term data area as well.

**Determining a shift-adjusted quantity**

This option known from MDE 7.2 is set by default in MOC. What this means is that postings that were generated as a result of a shift change are not considered when determining the quantity. All that is accounted for here is the exact moment when a machine status was set. If this moment is outside of the evaluation interval, in this case the output will be 0.

**The status assigned to RPA 12**

Status assigned to RPA 12 (typically the status for breaks and for shift-free times) are not selected.

**Status report detail application**

Tabular evaluation of the downtimes for a specific period of time and a certain number of workplaces. The result depends on the selection and therefore on the selection criteria made available on the selection panel.

The following data are available:

**Status category****Status**

Status number as per configuration

**Designation**

Designation of the status

**RPA**

Resource performance account number

**Status class**

Abbreviation of the status class

**Duration category****Duration**

Total time of all statuses determined within the evaluation period (i.e. not all downtimes within this evaluation period are compressed in this field).

%

Proportion of total status duration as compared to the total duration of disturbances shown as a percentage.

**Quantity category****Quantity**

Total number of statuses determined that were applicable during the evaluation period.

%

Proportion of the number of statuses to the total number shown as a percentage.

## **Comparison production - downtimes detail application**

In this detail application, the production time (green) accrued during the evaluation period and the sum total of all determined status times, not including RPA 11 and 12 (red) are compared in the form of a bar chart.

The presentation always relates to all of the displayed statuses (there is no highlighting option in this table).

## **Status hit list (durations) detail application**

The downtime durations for each status marked in the table are shown in the form of a graphic in the status hit list (durations) detail application.

In the combo box "Displayed series" you can define whether the durations displayed as a percentage should relate to the total number of downtimes (RPA 1-10) or to the total duration (RPA 1-11). If both options have been selected, both bars are displayed (for each status).

## **Status hit list (quantity) detail application**

The total number of status changes for each status marked in the table is shown in the form of a graphic in the status hit list (quantities) detail application.

As opposed to the previous detail applications, no further display options are available here.

## 4 Status Profile

### Summary

|                        |                                                                    |
|------------------------|--------------------------------------------------------------------|
| Menu                   | Operating facilities management → Status analyses → Status profile |
| Transaction code       | mstpf                                                              |
| Function authorization | mstpf                                                              |

### Usage

The status profile evaluates the statuses by shift, by day, by calendar week or by month over a certain period of time. The result depends on the selection and therefore on the selection criteria made available on the selection panel.

### Selection criteria

The application provides the following selection criteria:

#### Workplace

Search by machine/ by workplace. There is an option to search using wild cards.

#### Group

Search by workplaces/ machines that are assigned to the group that was entered. The selection is made using the field group in the workplace/ machine configuration. You can run a search using wildcards.

#### Cost center

Search by workplaces/ machines that are assigned to the cost center that was entered. You can run a search using wildcards.

#### Company

Search by workplaces/ machines that are assigned to the company that was entered. You can run a search using wildcards.

#### Responsibility area

This selection criterion references the responsibility area in the machine master data. Keep in mind that only machines are displayed that the user has also assigned responsibility areas to.

#### Report group

This selection criterion references the report groups. All machines or workplaces are displayed that are assigned to the selected report group.



**Short designation**

This selection criterion references the short name of the machines in the master data. All of the machines or workplaces are displayed that match the string that was entered. You can also run a search using wildcards.

**Designation**

This selection criterion references the name of the machines and workplaces in the machine's master data. At the bottom, only those machines are displayed that are identical to the string that was entered. You can also run a search using wildcards (placeholders \*) in this field.

**Status text**

Limits the selection to a certain status. Displayed in the combo box are the status texts that the selection can be filtered through.

**Inclusive status for RPA 11**

Accounts for the status for RPA 11 (usually, this is the "Production" status).

**Date**

The period of time from which data should be selected.

When selecting via shift(s), the shift date is evaluated, whereas when selecting by time, the selection is based on the start date. Please keep in mind that a selection by shift is only supported for ADE and MDE data, not for WRM data.

**Shift(s)/ time**

Selection by shift (only ADE and MDE events) or by time period. If no shift has been selected, all shifts are considered.

The two times each refer to the start or to the end of the date periods listed above.

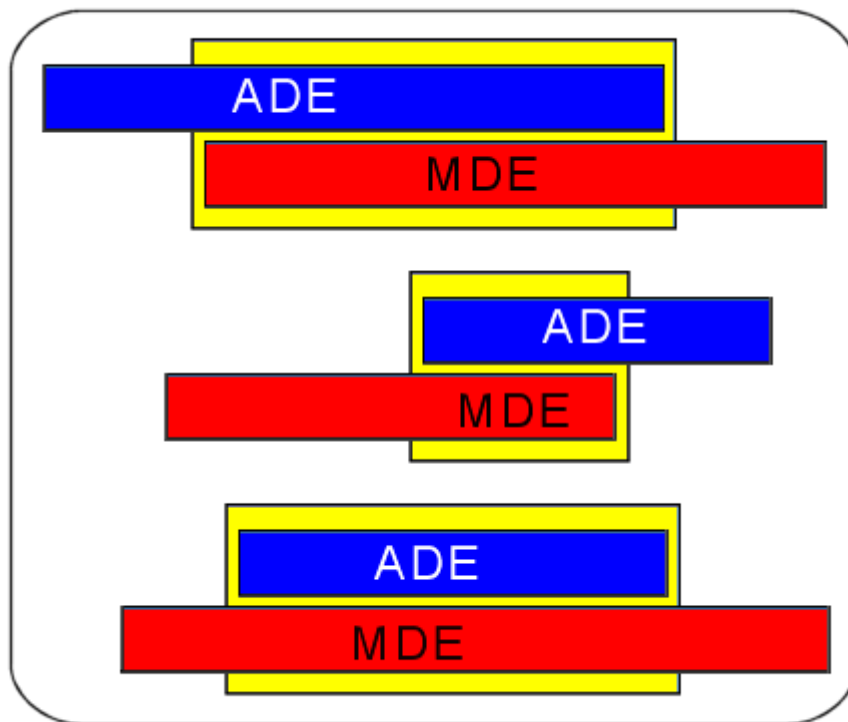
**Order/ article**

When you choose the option "Order", you must enter an order or an article.

For this kind of evaluation type, only the finished ADE postings are considered. If the order is currently still running on the machine, the time period between the last logon and now is not taken into account. As such, it is by all means possible that there are differences between the machine evaluation and the order-related evaluation. Only ADE postings are taken into account that have started during the evaluation period. If necessary, the selection period must be selected so that the ADE postings that are to be taken into account are within this selection period. For this order-related evaluation, MPDV recommends the shift-related selection option.



The following illustration shows an example of an overlapping of ADE and MDE postings. The ADE postings take priority in this evaluation type. The yellow areas illustrate the results of this evaluation. MDE quantities and durations are calculated on a pro rata basis to achieve the result.



If orders are logged on in parallel at the machine, for this evaluation type, the full machine time (yellow area) and quantity is assigned to each order. The fact that orders are run in parallel will not result in a proportionate calculation.

### Resource type/ resource

When choosing the option "Resource", a resource must be entered.

For this kind of evaluation type, here again only the finished postings are considered. For this evaluation type the resource postings take priority. The principle is the same as when running an evaluation by order.

### Additional selection notes

#### Long term data

If the selection period exceeds the period for the online data area, the system applies the implicit solution and selects the medium-term data area as well. Therefore, there is no need for an explicit activation in order to be able to access the medium-term data set.

### Determining a shift-adjusted quantity

This option known from MDE 7.2 is set by default in MOC. What this means is that postings that were generated as a result of a shift change are not considered when determining the quantity. All that is accounted for here is the exact moment when a machine status was set. If this moment is outside of the evaluation interval, in this case the output will be 0.



### Status profile detail application

Tabular presentation of the status with the following columns:

#### Shift date/ shift/ calendar week/ year

The durations in the statuses are displayed in groups by shift (i.e. shift date/ shift number). Calendar week, month and year (based on the shift date) are provided as additional sorting/ grouping criteria.

#### Status/ designation

Number and designation of the status. The status column is displayed in color as defined in the status text configuration.

#### RPA

Resource performance account number

#### Abbreviation

Abbreviation of the status class

#### Duration

Total time of all statuses within the evaluation period.

#### Quantity

Total number of statuses that was applicable during the evaluation period.

### Pivot table detail application

You can evaluate status based on additional criteria in the pivot table detail view.



The bar colors in the chart are set "arbitrarily" using a color chart defined internally.

## 5 Status Ranking List

### Overview

|                        |                                                                        |
|------------------------|------------------------------------------------------------------------|
| Menu                   | Production Facility Management → Status analyses → Status ranking list |
| Transaction code       | sthitl                                                                 |
| Function authorization | sthitl                                                                 |

### Purpose

The Status ranking list provides an overview of the most frequent or longest lasting statuses. The list indicates the duration and number of machine events collected as status. Also included in this overview are production statuses (statuses assigned to RPA 11) and break statuses.

There are two sorts of statuses: The machine/workplace status which is often referred to as "Downtime reason" or "Malfunction", and the further parallel statuses, e.g. program, operation type, operation mode or disturbances and production interruptions (depending on the license/project).

### Selection criteria

The application provides the following selection criteria:

#### Workplace

Workplaces/machines matching the criteria entered.

#### Group

Search by workplaces/ machines that are assigned to the group that was entered.

#### Date

Data should be selected from the entered period of time.

When selecting by shift(s), the shift date is evaluated, when selecting by time the selection is based on the start date. Please keep in mind that a selection by shift is only supported with BDE and MDE data, not with WRM data.



The display shows the evaluation of the selected period of time whether the data is already archived or not.

#### Shift/ time

Selection according to shifts (HYDRA-BDE and HYDRA-MDE events only) or according to periods. If no shift is selected, all shifts are integrated.

Both times refer respectively to the start or end of the date period specified above.

**Report group**

This selection criterion refers to the report groups. The application shows all workplaces/machines assigned to the selected report group.

**Responsibility area**

This selection criterion refers to the responsibility area in the machine master data. Please note that you may only view those machines you are authorized for by the responsibility area.

**Top**

Limits the number of statuses displayed for each selected machine to those with the longest duration. Pre-assignment: 5

**Status type**

Selection of status types that are included in the evaluation. By default, the machine status is available here; further status types are available depending on the license.

**Field descriptions****Resource**

Workplace/machine number

**Resource type**

For workplaces/machines always "MNR"

**Designation**

Designation of the workplace/machine

**Status, Status text**

Status number and status text of the status that was available. The status text is displayed in the status text color that was configured.

**Duration**

Duration indicating how long the current status was available.

**Total number**

Number of times a status was available.

**Status type**

Description of the status type a status belongs to. By default, the machine status is available here; further status types are available depending on the license or the project.

## 6 ABC Analysis

### Overview

|                        |                                                                 |
|------------------------|-----------------------------------------------------------------|
| Menu                   | Production Facility Management → Status analyses → ABC analysis |
| Transaction code       | stabc                                                           |
| Function authorization | stabc                                                           |

### Purpose

This report lists all malfunctions that occurred while the selected machine was running. The ABC analysis is intended to be a pure report on "Failures" = "Malfunctions". For this reason, the status "Production" is not evaluated.

The statuses are sorted according to the "Pareto Principle" - i.e sorted according to their size, summed up and classified - and classified as A, B and C depending on how long the status lasted. The threshold values are configurable.

### Selection criteria

The application provides the following selection criteria:

#### Workplace

Defines the workplace for which the ABC analysis is to be displayed.

#### Status type

Restricts the displayed error messages to one status type (depending on license or project).

#### Threshold value 1

Parameter used to set the ABC threshold values. For threshold value 1, the threshold is defined between the limits A and B. The predefined value is 50 %.

#### Threshold value 2

Parameter used to set the ABC threshold values. For threshold value 2, the threshold is defined between the limits B and C. The predefined value is 30 %.

#### Date from ...to (Shift / Time)

The error messages of the selected period of time are used.

### Field descriptions

#### Status

Status number. The coloring is based on the status text configuration.

**Status text**

Status name

**Status type**

The selection criteria restrict the displayed status type (depending on license or project). For example, the selection criteria provide the following status types:

- Machine status
- Malfunction
- Operation mode
- Operation state
- Program
- ...

**Status type designation**

Designation of the active status type

**Duration, %**

Total status duration indicating how long the status was active at the machine and percentage of the total duration.

**Quantity, %**

Number indicating how often this status was active at the machine and percentage of the total number.

**Shift**

Shift number indicating the shift when the status was active.

**Shift start / End of shift**

Beginning and end of shift during which the status was active.

**Detail application *ABC analysis***

The detail application *ABC analyses* provides a sum total of all accrued durations and displays the number of individual postings included. The data is classified in the three classes A, B and C. The classification is based on the percentages referring to the total duration. The values are totaled according to the "Pareto principle", i.e. the individual rows are sorted by their size in descending order into the classes A to C and added to the class until the total sum exceeds the threshold value (to be more precise: threshold value specified 100 %). Then the next class is filled.

## **Detail application *Individual listing***

If you select a row in the ABC analysis, the *Individual listing* shows the individual rows included in the selected row.



## 7 Minor/Major Stops

### Overview

|                        |                                                                      |
|------------------------|----------------------------------------------------------------------|
| Menu                   | Production Facility Management → Status analyses → Minor/major stops |
| Transaction code       | minmaj                                                               |
| Function authorization | minmaj                                                               |

### Purpose

This report shows production interruptions for a selected machine. Subject to their duration, interruptions are classified as minor and major stops. Minor stops are shorter interruptions. They do not affect production processes, as they are compensated by buffer times. But if a "minor stop" exceeds a threshold value previously specified, it becomes a "major stop".

In the lists of the minor and major stops, all stops with the same cause or the same text are summarized in one position. The single causes are listed by expanding one cause. The last stop that occurred is shown at the top of the list.

### Selection criteria

The application provides the following selection criteria:

#### Workplace

Defines the workplace for which the stops are to be displayed.

#### Date from ...to (Shift / Time)

The stops of the selected period of time are used.

#### Status type

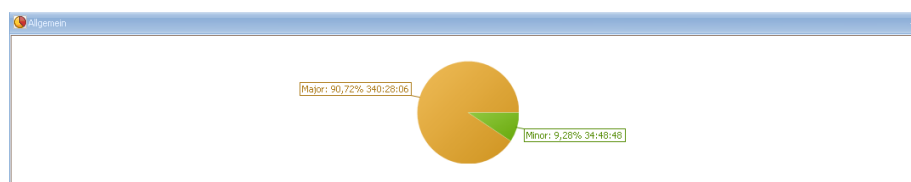
Restricts the displayed stops to one defined status type.

If you did not select a status type, the status type *Machine status* (MST) is selected by default.

#### Minor/major thresholds

This threshold specifies, if a stop is a minor or a major stop. The predefined value is 5 minutes.

### Detail application *General*



Graphic presentation of the total time and how the time is distributed in percent to minor and major stops.

## Detail application *Minor report*

The *Minor report* shows the selected stops of the selected machine. The duration of the displayed stops is shorter than the minor/major threshold entered.

Minor-Report

Ziehen Sie eine Spalte hierher, um nach dieser zu gruppieren

| Status | Status   | Status | Status | Dauer    | %    | Anzahl | %     |
|--------|----------|--------|--------|----------|------|--------|-------|
| Status | Status   | Status | Status | Dauer    | %    | Anzahl | %     |
| 7,0    | Break    | MST    |        | 10:57:35 | 2,92 | 901    | 30,85 |
| 4,0    | Cleaning | MST    |        | 9:21:47  | 2,49 | 867    | 29,68 |
| 103,0  | Prepared | MST    |        | 6:59:36  | 1,86 | 476    | 16,30 |
| 105,0  | Tailback | MST    |        | 4:02:59  | 1,08 | 258    | 8,83  |

Einzelauflistung

Ziehen Sie eine Spalte hierher, um nach dieser zu gruppieren

| Status | Status   | Status | Status | Dauer             | Ende              | Dauer   | Schicht | Schichtbeginn   | Schichtende    |
|--------|----------|--------|--------|-------------------|-------------------|---------|---------|-----------------|----------------|
| Status | Status   | Status | Status | Start             | Ende              | Dauer   | Schicht | Schichtbeginn   | Schichtende    |
| 105,0  | Tailback | MST    |        | 02.07.2010 00:... | 02.07.2010 00:... | 0:01:20 | 3       | 01.07.2010 2... | 02.07.2010 ... |
| 105,0  | Tailback | MST    |        | 01.07.2010 23:... | 01.07.2010 23:... | 0:01:59 | 3       | 01.07.2010 2... | 02.07.2010 ... |
| 105,0  | Tailback | MST    |        | 01.07.2010 23:... | 01.07.2010 23:... | 0:00:33 | 3       | 01.07.2010 2... | 02.07.2010 ... |
| 105,0  | Tailback | MST    |        | 01.07.2010 23:... | 01.07.2010 23:... | 0:01:35 | 3       | 01.07.2010 2... | 02.07.2010 ... |
| 105,0  | Tailback | MST    |        | 01.07.2010 23:... | 01.07.2010 23:... | 0:02:19 | 3       | 01.07.2010 2... | 02.07.2010 ... |

34:48:48 9,28 2.885 98,77

## Detail application *Major report*

Major-Report

Ziehen Sie eine Spalte hierher, um nach dieser zu gruppieren

| Status | Status               | Status | Status | Dauer     | %     | Anzahl | %    |
|--------|----------------------|--------|--------|-----------|-------|--------|------|
| Status | Status               | Status | Status | Dauer     | %     | Anzahl | %    |
| 7,0    | Break                | MST    |        | 209:10:18 | 55,74 | 11     | 0,38 |
| 4,0    | Cleaning             | MST    |        | 71:26:46  | 19,04 | 15     | 0,51 |
| 110,0  | Aborting             | MST    |        | 51:07:29  | 13,62 | 1      | 0,03 |
| 105,0  | Tailback             | MST    |        | 3:54:34   | 1,04  | 3      | 0,10 |
| 9,0    | Operating (start up) | MST    |        | 2:32:13   | 0,68  | 2      | 0,07 |

Einzelauflistung

Ziehen Sie eine Spalte hierher, um nach dieser zu gruppieren

| Status | Status               | Status | Status | Dauer             | Ende              | Dauer   | Schicht | Schichtbeginn   | Schichtende    |
|--------|----------------------|--------|--------|-------------------|-------------------|---------|---------|-----------------|----------------|
| Status | Status               | Status | Status | Start             | Ende              | Dauer   | Schicht | Schichtbeginn   | Schichtende    |
| 9,0    | Operating (start up) | MST    |        | 06.07.2010 13:... | 06.07.2010 14:... | 0:19:19 | 1       | 06.07.2010 0... | 06.07.2010 ... |
| 9,0    | Operating (start up) | MST    |        | 05.07.2010 13:... | 05.07.2010 15:... | 2:12:54 | 1       | 05.07.2010 0... | 05.07.2010 ... |

2:32:13

|       |                     |     |  |         |      |   |      |
|-------|---------------------|-----|--|---------|------|---|------|
| 103,0 | Prepared            | MST |  | 2:01:34 | 0,54 | 3 | 0,10 |
| 8,0   | Starting (start up) | MST |  | 0:15:12 | 0,07 | 1 | 0,03 |

340:28:06 90,72 36 1,23

The *Major report* shows the selected stops of the selected machine. The duration of the displayed stops is longer than the minor/major threshold entered.

## Field descriptions

### Status

Status number and also cause of the accrued stop.

### Status text

Status text of the status.

### Status type

The selection criteria restrict the displayed status type. For example, the selection criteria can provide the following status types (depending on license or project):

- Machine status

- Malfunction
- Operation mode
- Operation state
- Program
- Sequencer A - module
- Sequencer A - programs
- Sequencer A - steps
- Sequencer B - module
- Sequencer B - programs
- Sequencer B - steps
- Sequencer gen. - status

**Status type designation**

Designation of the active status type

**Start / End**

Beginning and end of stop

**Duration, %**

Total stop duration indicating how long the stop was active at the machine and percentage of total duration.

**Quantity, %**

Total status number indicating how often this status was active and percentage of the total number.

**Shift**

Shift number indicating the shift when the stop occurred.

**Shift start / End of shift**

Beginning and end of shift during which the stop occurred.

## 8 Status Class Report

### Summary

|                        |                                                                         |
|------------------------|-------------------------------------------------------------------------|
| Menu                   | Operating facilities management → Status analyses → Status class report |
| Transaction code       | stclrp                                                                  |
| Function authorization | stclrp                                                                  |

### Usage

The evaluation provides status information for a specific period of time and a certain number of workplaces. In it, the statuses are consolidated into status classes that were assigned in a previous step. The application provides the user with the ability to run an analysis of the booked status classes.

### Selection criteria

The application provides the following selection criteria:

#### Workplace

Search by machine/ by workplace. You can run a search using wildcards.

#### Group

Search by workplaces/ machines that are assigned to the group that was entered. The selection is made using the field Group in the workplace/ machine configuration. You can run a search using wildcards.

#### Cost center

Search by workplaces/ machines that are assigned to the cost center that was entered. You can run a search using wildcards.

#### Company

Search by workplaces/ machines that are assigned to the company that was entered. You can run a search using wildcards.

#### Responsibility area

This selection criterion references the responsibility area in the machine master data. Keep in mind that only machines are displayed that the user has also assigned responsibility areas to.

#### Report group

This selection criterion references the report groups. All machines or workplaces are displayed that are assigned to the selected report group.

**Short designation**

This selection criterion references the short name of the machines in the master data. All of the machines or workplaces are displayed that match the string that was entered. You can also run a search using wildcards.

**Designation**

This selection criterion references the name of the machines and workplaces in the machine's master data. At the bottom, only those machines are displayed that are identical to the string that was entered. You can also run a search using wildcards (placeholders \*) in this field.

**Date**

The period of time from which data should be selected.

When selecting via shift(s), the shift date is evaluated, while when selecting by time the selection is based on the start date. Please keep in mind that a selection by shift is only supported for BDE and MDE data, not for WRM data.

**Shift(s)/ time**

Selection by shift (only BDE and MDE events) or by time period. If no shift has been selected, all shifts are considered.

The two times each refer to the start or to the end of the date periods listed above.

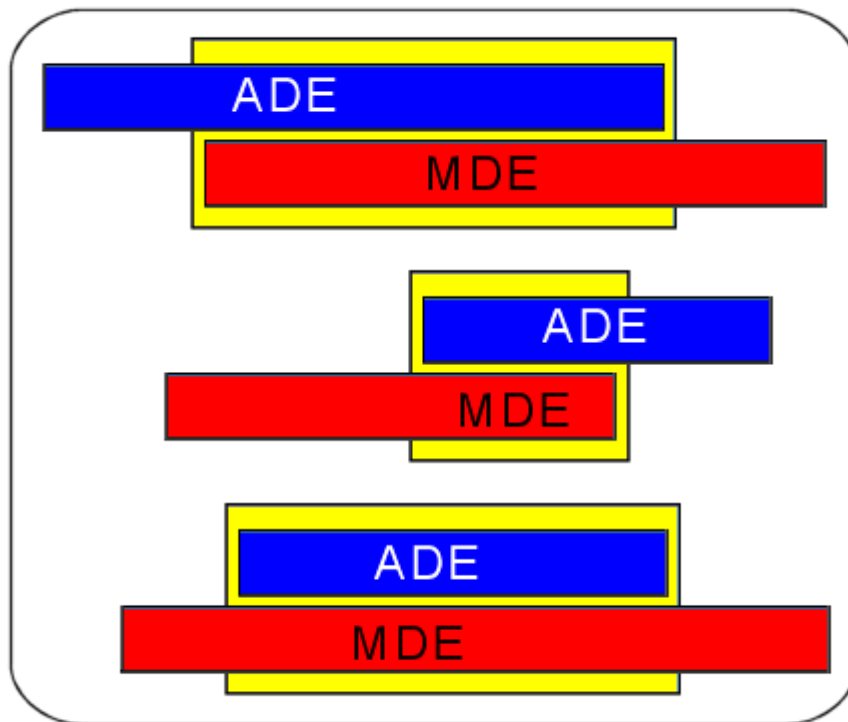
**Order/ article**

When you choose the option "Order", you must enter an order or an article.

For this kind of evaluation type, only the finished BDE postings are considered. If the order is currently still running on the machine, the time period between the last logon and now is not taken into account. As such, it is by all means possible that there are differences between the machine evaluation and the order-related evaluation. Only BDE postings are taken into account that have started during the evaluation period. If necessary, the selection period must be selected so that the BDE postings that are to be taken into account are within this selection period. For this order-related evaluation, MPDV recommends the shift-related selection option.



The following illustration shows an example of an overlapping of BDE and MDE postings. The BDE postings take priority in this evaluation type. The yellow areas illustrate the results of this evaluation. MDE quantities and durations are calculated on a pro rata basis to achieve the result.



If orders are logged on in parallel at the machine, for this evaluation type, the full machine time (yellow area) and quantity is assigned to each order. The fact that orders are run in parallel will not result in a proportionate calculation.

### Resource type/ resource

When choosing the option "Resource", a resource must be entered.

For this kind of evaluation type, here again only the finished postings are considered. For this evaluation type, only the resource postings take priority. The principle is the same as when running an evaluation by order.

## Additional selection notes

### Long term data

If the selection period exceeds the period for the online data area, the system applies the implicit solution and selects the medium-term data area as well. Therefore, there is no need for an explicit activation in order to be able to access the medium-term data set.

### Determining a shift-adjusted quantity

This option known from MDE 7.2 is set by default in MOC. What this means is that postings that were generated as a result of a shift change are not considered when determining the quantity. All that is accounted for here is the exact moment when a machine status was set. If this moment is outside of the evaluation interval, in this case the output will be 0.



## Status class report detail application

Tabular presentation of the status class reports

### Status class

Status class number

### Abbreviation

Abbreviation of the status class

### Designation

Description of the status class

### Duration

Total time of all statuses within the evaluation period, accumulated to status classes.

%

Proportion of the total duration of the status classes.

### Quantity

Total number of all statuses within the evaluation period, accumulated to status classes.

%

Proportion of the number of statuses to the total number of all statuses.

## Status class hit list (durations) detail application

Displayed in a bar chart in the status class hit list (durations) detail application are the durations posted to status classes. Displayed here are the status classes that were highlighted in the tabular detail application status class report. The red bars are shown in descending order by duration.

## Status class hit list (quantities) detail application

Displayed in a bar chart in the status class hit list (quantities) detail application are the number of status classes. Displayed here are the status classes that were highlighted in the tabular detail application status class report. The red bars are shown in descending order by quantity.

## 9 Status Class Profile

### Summary

|                        |                                                                          |
|------------------------|--------------------------------------------------------------------------|
| Menu                   | Operating facilities management → Status analyses → Status class profile |
| Transaction code       | stclpf                                                                   |
| Function authorization | stclpf                                                                   |

### Usage

The status class profile evaluates the status classes by shift, by day, by calendar week or by month over a certain period of time. The result depends on the selection and therefore on the selection criteria made available on the selection panel.

### Selection criteria

The application provides the following selection criteria:

#### Workplace

Search by machine/ by workplace. You can run a search using wildcards.

#### Group

Search by workplaces/ machines that are assigned to the group that was entered. The selection is made using the field Group in the workplace/ machine configuration. You can run a search using wildcards.

#### Cost center

Search by workplaces/ machines that are assigned to the cost center that was entered. You can run a search using wildcards.

#### Company

Search by workplaces/ machines that are assigned to the company that was entered. You can run a search using wildcards.

#### Responsibility area

This selection criterion references the responsibility area in the machine master data. Keep in mind that only machines are displayed that the user has also assigned responsibility areas to.

#### Report group

This selection criterion references the report groups. All machines or workplaces are displayed that are assigned to the selected report group.



**Short designation**

This selection criterion references the short name of the machines in the master data. All of the machines or workplaces are displayed that match the string that was entered. You can also run a search using wildcards.

**Designation**

This selection criterion references the short name of the machines and workplaces in the machine's master data. The machines or workplaces are displayed that are identical to the string that was entered. You can also run a search using wildcards (placeholders \*) in this field.

**Date**

The period of time from which data should be selected.

When selecting via shift(s), the shift date is evaluated, whereas when selecting by time, the selection is based on the start date. Please keep in mind that a selection by shift is only supported for BDE and MDE data, not for WRM data.

**Shift(s)/ time**

Selection by shift (only BDE and MDE events) or by time period. If no shift has been selected, all shifts are considered.

The two times each refer to the start or to the end of the date periods listed above.

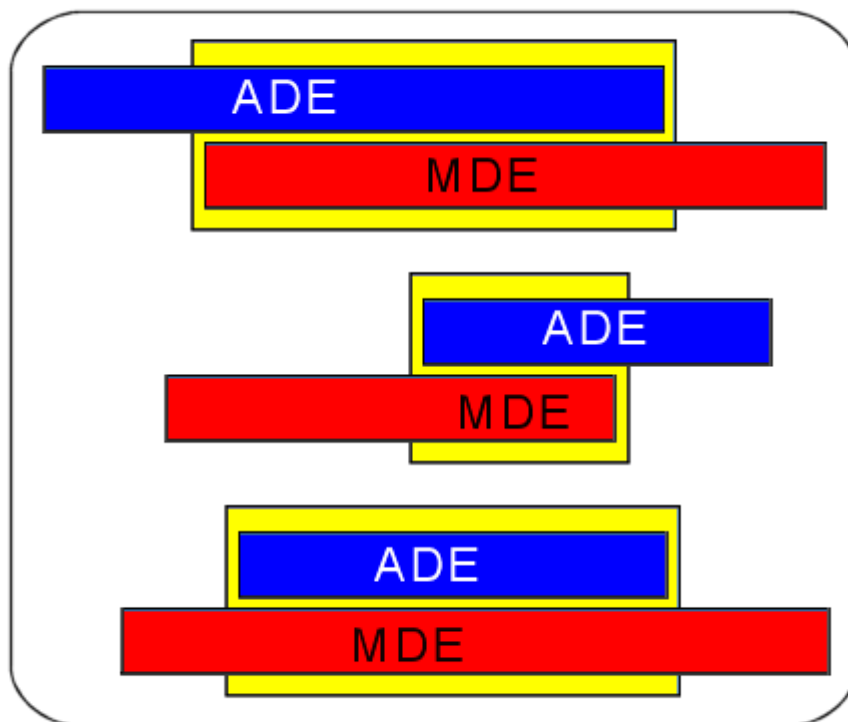
**Order/ article**

When you choose the option "Order", you must enter an order or an article.

For this kind of evaluation type, only the finished BDE postings are considered. If the order is currently still running on the machine, the time period between the last log on and now is not taken into account. As such, it is by all means possible that there are differences between the machine evaluation and the order-related evaluation. Only BDE postings are taken into account that have started during the evaluation period. If necessary, the selection period must be selected so that the BDE postings that are to be taken into account are within this selection period. For this order-related evaluation, MPDV recommends the shift-related selection option.



The following illustration shows an example of an overlapping of BDE and MDE postings. The BDE postings take priority in this evaluation type. The yellow areas illustrate the results of this evaluation. MDE quantities and durations are calculated on a pro rata basis to achieve the result.



If orders are logged on in parallel at the machine, for this evaluation type, the full machine time (yellow area) and quantity is assigned to each order. The fact that orders are run in parallel will not result in a proportionate calculation.

### Resource type/ resource

When choosing the option "Resource", a resource must be entered.

For this kind of evaluation type, here again only the finished postings are considered. For this evaluation type, only the resource postings take priority. The principle is the same as when running an evaluation by order.

### Additional selection notes

#### Long term data

If the selection period exceeds the period for the online data area, the system applies the implicit solution and selects the medium-term data area as well. Therefore, there is no need for an explicit activation in order to be able to access the medium-term data set.

**Determining a shift-adjusted quantity**

This option known from MDE 7.2 is set by default in MOC. What this means is that postings that were generated as a result of a shift change are not considered when determining the quantity. All that is accounted for here is the exact moment when a machine status was set. If this moment is outside of the evaluation interval, in this case the output will be 0.

**Status class profile detail application**

Tabular presentation of the status class profile with the following columns:

**Shift date/ shift/ calendar week/ year**

The durations of the status classes are displayed per shift (i.e. shift date/ shift number). Calendar week, month and year (based on the shift date) are provided as additional sorting/ grouping criteria.

**Status class/ abbreviation/ designation**

Number, abbreviation and designation used for the status class

**Duration**

Time duration in which the status class was created/ set within the evaluation period.

**Quantity**

Number of times that the status class was created/ set within the evaluation period.

**Pivot table detail application**

You can evaluate status classes based on additional criteria in the pivot table detail view.



The bar colors in the chart are set "arbitrarily" using a color chart defined internally.

## 10 RPA Report

### Overview

|                        |                                                               |
|------------------------|---------------------------------------------------------------|
| Menu                   | Production facility management → Status analyses → RPA report |
| Transaction code       | rpap                                                          |
| Function authorization | rpap                                                          |

### Purpose

The RPA report provides status information of workplaces/machines over a specified time and for a specified number of workplaces. For the report, the workplace/machine statuses are assigned to resource performance accounts where they are totaled.

### Selection criteria

The application provides the following selection criteria:

#### Workplace

Selection by machine/workplace. You can use wildcards.

#### Group

Selection by workplaces/machines that are assigned to the machine group specified. The selection is made using the field *Group* in the *Workplace/machine configuration*. You can use wildcards.

#### Cost center

Selection by workplaces/machines that are assigned to the cost center specified. You can use wildcards.

#### Company

Selection by workplaces/machines that are assigned to the company specified. You can use wildcards.

#### Responsibility area

This selection criterion refers to the responsibility area stored in the machine master data. Note: The user can only view those machines that are included in the responsibility areas assigned to the user.

#### Report group

This selection criterion refers to the report groups. The application shows all workplaces/machines assigned to the selected report group.

**Short name**

This selection criterion refers to the short name of machines in the master data. The application shows all machines or workplaces matching the entered character string. You can also use wildcards.

**Designation**

This field refers to the designation/name of machines and workplaces defined in the machine master data (in HYDRA: comment). The application only shows the machines matching the character string specified. You can also use wildcards (placeholders \*).

**Including status for RPA 11**

The status assigned to RPA 11 is integrated in the report (usually status "production").

**Date**

The data included in the period of time specified is used.

If you perform the selection using shift(s), the shift date is evaluated, if you use the time for selection, the selection is based on the start date. Note: a selection by shift is only supported with BDE and MDE data, not with WRM data.

**Shift(s)/time**

Selection by shifts (HYDRA-BDE and HYDRA-MDE events only) or using a period of time. If no shift is selected, all shifts are used.

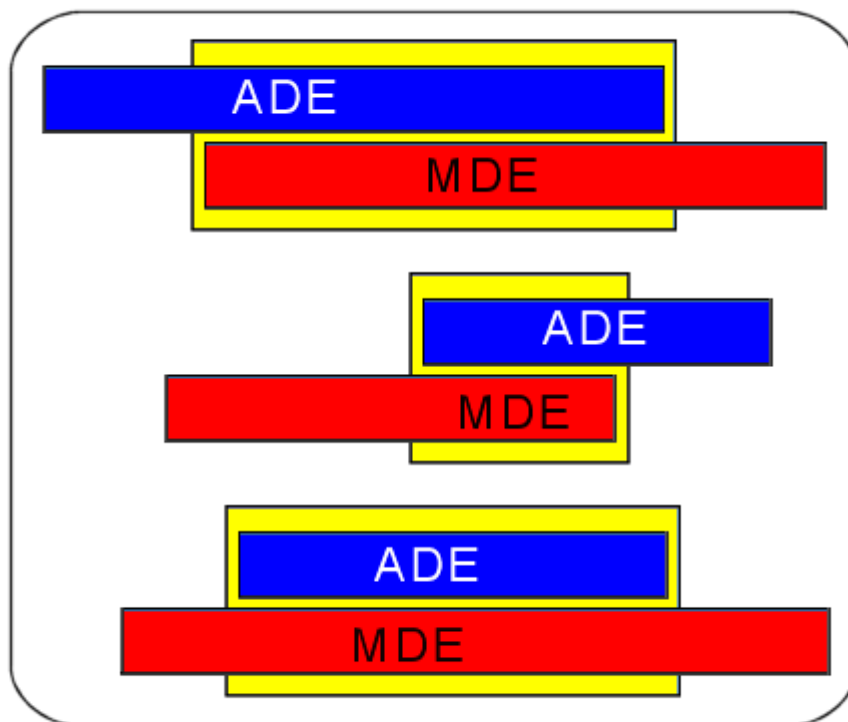
Both times refer to the beginning or end of the date period specified above.

**Order/article**

If you select the option *Order*, you must specify an order or an article.

With this selection type, only completed BDE postings are used. If the order is still running at the machine, the system does not integrate the time period between the last logon and now. It is therefore possible that there is a difference between the machine evaluation and the order-related evaluation. The system only uses BDE postings that start in the evaluation period. If required, you must specify the selection period so that the required BDE postings are actually included in this period. For this order-related evaluation, MPDV recommends to select data by shift.

The illustration below shows an example of how BDE and MDE postings can overlap. The BDE postings take priority with this evaluation type. The yellow fields show the result of this evaluation. MDE quantities and times are used proportionately to calculate the result.



If several orders are logged on to the machine at the same time, this evaluation type assigns the complete machine time and number of pieces to each of the orders (yellow field). Times and quantities are not assigned proportionately when orders are logged on in parallel.

#### Resource type/resource

If you select the option *Resource*, you must specify a resource.

Also with this selection type, only completed postings are used. With this evaluation type, the resource postings take priority. The principle is the same as for evaluations by order.

### Additional notes on the selection

#### Long-term data

If the selection period exceeds the period of time of the online data area, the system implicitly selects the data of the medium-term data area. You need not explicitly activate the access to the medium-term data area.

#### Using quantities during time of shift only

This option known from MDE 7.2 is set by default in MOC. With this option, the postings created during shift change are not used when the quantity is identified. The machine status must have been set and only then the quantities produced are used. If this moment is outside of the evaluation interval, then the output is 0.

## **Detail application *RPA report***

### **RPA / Abbrev. / Designation**

Number, abbreviation and name of the resource performance account.

### **Duration**

Time that the status lasted/was set that is assigned to this RPA.

%

Share of time in the total time.

### **Quantity**

Number of times that the status assigned to this RPA was available/set.

%

Share of the number of times in the total number of times.

## **Detail application *Duration***

The detail application *Duration* displays the durations of the resource performance accounts in a bar chart. The durations are sorted and displayed in descending order. The different resource performance accounts are colored according to the default definition.

## **Detail application *Quantity***

The detail application *Quantity* displays the number of times that a status was available at the selected machine. The statuses are displayed according to the RPA in a bar chart. The number of times are sorted and displayed in descending order. The different resource performance accounts are colored according to the default definition.

## 11 PRA Profile

### Summary

|                        |                                                                 |
|------------------------|-----------------------------------------------------------------|
| Menu                   | Operating facilities management → Status analyses → RPA profile |
| Transaction code       | rpap                                                            |
| Function authorization | rpap                                                            |

### Usage

The RPA profile provides status information for a specific period of time and a certain number of workplaces. In it, the status of resource performance accounts that were assigned in a previous step are consolidated and displayed by shift.

### Selection criteria

The application provides the following selection criteria:

#### Workplace

Search by machine/ by workplace You can also run a search using wildcards.

#### Group

Search by workplaces/ machines that are assigned to the group that was entered. The selection is made using the field Group in the workplace/ machine configuration. You can also run a search using wildcards.

#### Cost center

Search by workplaces/ machines that are assigned to the cost center that was entered. You can also run a search using wildcards.

#### Company

Search by workplaces/ machines that are assigned to the company that was entered. You can also run a search using wildcards.

#### Responsibility area

This selection criterion references the responsibility area in the machine master data. Keep in mind that only machines are displayed that the user has also assigned responsibility areas to.

#### Report group

This selection criterion references the report groups. All machines or workplaces are displayed that are assigned to the selected report group.



**Short name**

This selection criterion references the short name of the machines in the master data. All of the machines or workplaces are displayed that match the string that was entered. You can also run a search using wildcards.

**Designation**

This field references the designation (in HYDRA: comment) used for the machines and workplaces in the machine master data. At the bottom, only those machines are displayed that are identical to the string that was entered. There is also the option to use wild cards (placeholders \*) in this field.

**Inclusive status for RPA 11**

Accounts for the status for RPA 11 (usually, this is the "Production" status).

**Date**

The period of time from which data should be selected.

When selecting via shift(s), the shift date is evaluated, while when selecting by time the selection is based on the start date. Please keep in mind that a selection by shift is only supported for BDE and MDE data, not for WRM data.

**Shift(s)/ time**

Selection by shift (only BDE and MDE events) or by time period. If no shift has been selected, all shifts are considered.

The two times each refer to the start or to the end of the date periods listed above.

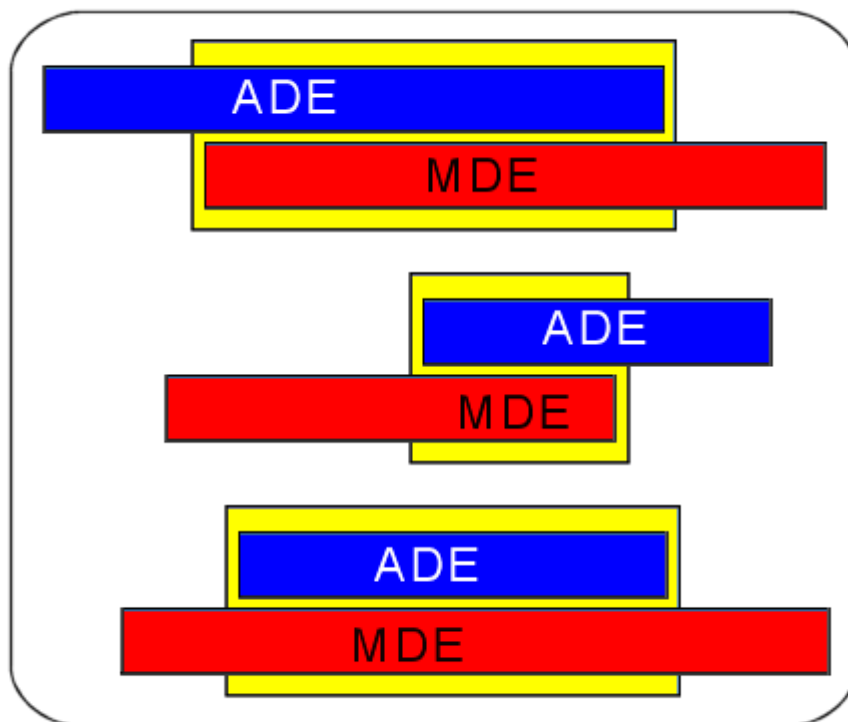
**Order/ article**

When you choose the option "Order", you must enter an order or an article.

For this kind of evaluation type, only the finished BDE postings are considered. If the order is currently still running on the machine, the time period between the last log on and now is not taken into account. As such, it is by all means possible that there are differences between the machine evaluation and the order-related evaluation. Only BDE postings are taken into account that have started during the evaluation period. If necessary, the selection period must be selected so that the BDE postings that are to be taken into account are within this selection period. For this order-related evaluation, MPDV recommends the shift-related selection option.



The following illustration shows an example of an overlapping of BDE and MDE postings. The BDE postings take priority in this evaluation type. The yellow areas illustrate the results of this evaluation. MDE quantities and durations are calculated on a pro rata basis to achieve the result.



If orders are logged on in parallel at the machine, for this evaluation type, the full machine time (yellow area) and quantity is assigned to each order. The fact that orders are run in parallel will not result in a proportionate calculation.

### Resource type/ resource

When choosing the option "Resource", a resource must be entered.

For this kind of evaluation type, here again only the finished postings are considered. For this evaluation type, only the resource postings take priority. The principle is the same as when running an evaluation by order.

## Additional selection notes

### Long term data

If the selection period exceeds the period for the online data area, the system applies the implicit solution and selects the medium-term data area as well. Therefore, there is no need for an explicit activation in order to be able to access the medium-term data set.

### Determining a shift-adjusted quantity

This option known from MDE 7.2 is set by default in MOC. What this means is that postings that were generated as a result of a shift change are not considered when determining the quantity. All that is accounted for here is the exact moment when a machine status was set. If this moment is outside of the evaluation interval, in this case the output will be 0.



## Overview of detail application

### Shift date/ shift/ calendar week/ year

The durations in the resource performance accounts are displayed in groups by shift (i.e. shift date/ shift number).

Calendar week and year are displayed as a possible sorting/ grouping criterion. The calendar week and the year match the shift date.

### RPA/ abbreviation/ designation

When delivered from the factory, the durations set for the resource performance accounts 1-10 are listed in the pool of columns. They are stored there with their abbreviation and can be shown as needed.

### Duration

Time duration in which the status was created/ set within the evaluation period.

### Quantity

Number of times that the status was created/ set within the evaluation period.

## Duration detail application

In the duration detail application, the resource performance account durations are displayed in the form of a stacked bar chart. They are displayed in chronological form (X axis) or accumulated to shift date (Y axis). The color code for each of the resource performance accounts is in accordance with the standard definitions.

## Quantity detail application

In the quantity detail application, the number of statuses for the selected machines are displayed in the form of a stacked bar chart, broken down by RPA. They are displayed in chronological form (X axis) or accumulated to shift date (Y axis). The color code for each of the resource performance accounts is in accordance with the standard definitions.

## 12 Status Analysis

### Summary

|                        |                                                                     |
|------------------------|---------------------------------------------------------------------|
| Menu                   | Operating facilities management → Status analyses → Status analysis |
| Transaction code       | stata                                                               |
| Function authorization | stata                                                               |

### Usage

Because of how flexible it is, the status analysis answers all questions relating to downtimes, malfunction reasons and production times, information that is useful for the shift foreman and team leaders, from the production controller to the production manager.

The recorded production times represent the basic data made available by the status analysis. The status postings are provided here in the finest detail and using the pivot table function, they can be compiled to generate informative reports.

### Selection criteria

The application provides the following selection criteria:

#### Report group

This selection criterion references the report groups. All machines or workplaces are displayed that are assigned to the selected report group.

#### Cost center

Search by workplaces/ machines that are assigned to the cost center that was entered. You can search using wild cards.

#### Workplace

Search by machine/ by workplace. You can search using wild cards.

#### Responsibility area

This selection criterion references the responsibility area in the machine master data. Keep in mind that only machines are displayed that the user has also assigned responsibility areas to.

#### Company

Search by workplaces/ machines that are assigned to the company that was entered. You can search using wild cards.

#### Group

Search by workplaces/ machines that are assigned to the group that was entered.

**Date**

The period of time from which data should be selected.

When selecting via shift(s), the shift date is evaluated, whereas when selecting by time the selection is based on the start date. Please keep in mind that a selection by shift is only supported for ADE and MDE data, not for WRM data.

**Shift/ time**

Selection by shift or by time period. If no shift has been selected, all shifts are considered.

The two times each refer to the start or to the end of the date periods listed above.

**Additional selection notes****Long term data**

If the selection period exceeds the period for the online data area, the system applies the implicit solution and selects the medium-term data area as well. Therefore, there is no need for an explicit activation in order to be able to access the medium-term data set.

**Determining a shift-adjusted quantity**

This option known from MDE 7.2 is set by default in MOC. What this means is that postings that were generated as a result of a shift change are not considered when determining the quantity. All that is accounted for here is the exact moment when a machine status was set. If this moment is outside of the evaluation interval, in this case the output will be 0.

**Status analysis detail application**

The results that were found are displayed in this detail application in tabular form. The results answer the question: When was a certain status recorded for which machine and for how long?

The following columns are shown, among others:

**Workplace/ short designation**

Number and short name of the workplace

**Beginning/ end**

Beginning period or ending period of the status respectively.

**Shift date/ shift number**

Shift date and shift in which the status applied.

**Status/ status text**

Number and designation of the active status

**Duration**

Duration of the status

### Quantity

The Quantity field reflects how often a status applied. Because only one status is displayed here at a time, generally a 1 is displayed as the quantity.

A 0 is displayed if the status was set due to an automatic status change that took place during a shift change and it is no different than the previous status (before the shift change).

### Pivot table detail application

Available in the detail application "Pivot table" are data found that are used for a pivot analysis. Functions such as those known from Microsoft Excel® can be used for this purpose.



The colors used in the graphic do not depend on the colors used for status configuration.